



# Skole

## How the system works

A short, plain-language guide: who uses the school system, what each person can do, how the data fits together and how it runs.

**Live site:** <https://skoleph.web.app>

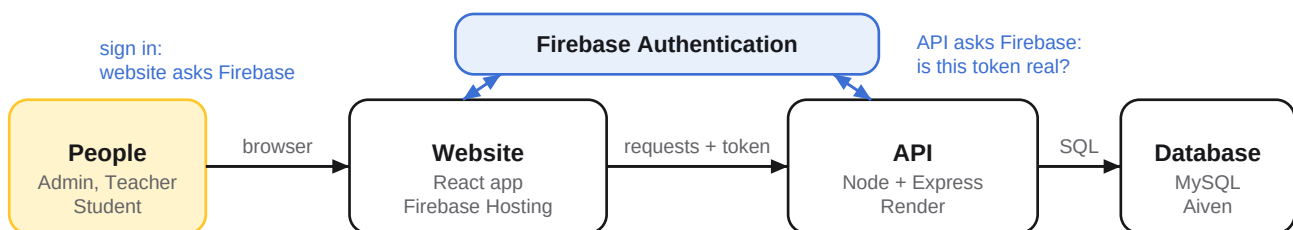
**Source code:** <https://github.com/Lianhahaha/School-Management-System>

**Demo sign-in:** admin@school.test, teacher1@school.test or student1@school.test, password Password123!

# 1. The big picture

Skole is a web application for running a school. It works in any browser, with nothing to install. Administrators, teachers and students sign in, and each sees only what belongs to them.

| Piece           | Job                                                                            | Runs on                 |
|-----------------|--------------------------------------------------------------------------------|-------------------------|
| <b>Website</b>  | The screens you click (React).                                                 | Firebase Hosting        |
| <b>API</b>      | The back office: checks who you are, applies the rules, reads and writes data. | Render                  |
| <b>Database</b> | Stores every school record.                                                    | Aiven MySQL             |
| <b>Sign-in</b>  | Knows email and password and proves who you are. Holds no school data.         | Firebase Authentication |



Every school record lives in MySQL. Firebase only knows who you are (email + password).

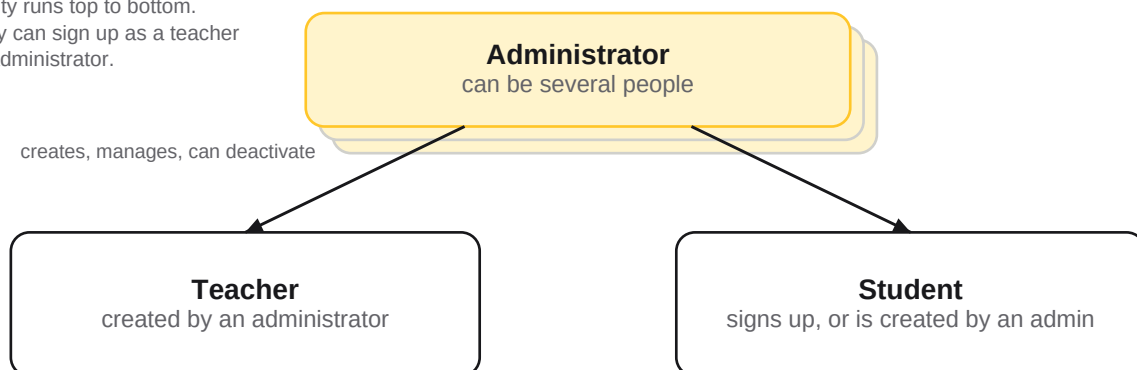
## Why split it this way

The website never touches the database. Every action goes through the API, which checks who you are and what you may do first, so editing a web address cannot reveal anyone else's data.

# 2. People and authority

There are three roles. A role is chosen when the account is created and never changes.

Authority runs top to bottom.  
Nobody can sign up as a teacher or an administrator.



**Administrator:** the highest authority. Manages all accounts and all school data. **There can be several** (the demo has two). Nobody can sign up as one: an existing administrator creates it, and the first comes from the setup command.

**Teacher:** works with the classes they are assigned to: marks attendance, creates assessments and enters grades for their own lessons, posts announcements to their classes.

**Student:** sees their own class, timetable, attendance, grades and announcements. Read-only, except their own contact details.

### The administrator is protected from mistakes

You cannot deactivate or delete **your own** account. The **last active administrator** can never be removed, even if two try at once. An account with school records can be deactivated but not deleted. Role and email cannot be edited after creation.

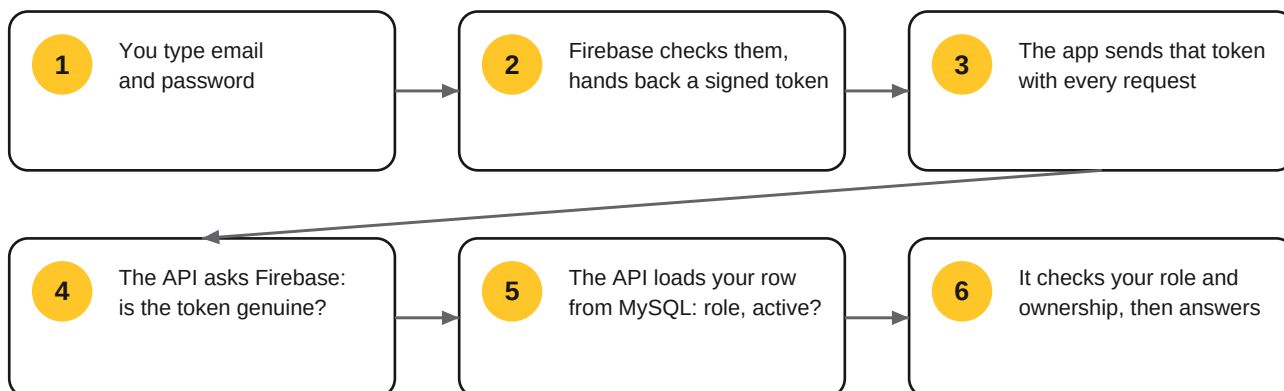
| Capability                                                            | Admin  | Teacher        | Student    |
|-----------------------------------------------------------------------|--------|----------------|------------|
| Create and deactivate accounts                                        | Yes    | No             | No         |
| Subjects, classes, teacher assignment, enrollments, timetable: change | Yes    | No             | No         |
| Student profiles: view                                                | All    | Their classes  | Themselves |
| Attendance: mark                                                      | Yes    | Own lessons    | No         |
| Assessments and grades: enter                                         | Yes    | Own lessons    | No         |
| Attendance and grades: view                                           | All    | Own lessons    | Own record |
| Announcements: post                                                   | Anyone | To own classes | No         |
| Edit own contact details                                              | Yes    | Yes            | Yes        |

## 3. Accounts and sign-in

| Role                 | How the account is made                                                                                                                                    |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Administrator</b> | Created by an administrator (Users, Create user). The first one comes from the setup command.                                                              |
| <b>Teacher</b>       | Created by an administrator, who sets a temporary password and gives it to the teacher. Teachers cannot sign up themselves.                                |
| <b>Student</b>       | Signs up on the public Create account page, or an administrator creates it. With no class yet, only the profile shows until an administrator enrolls them. |

**There is no "promote" button.** A role decides which records an account owns, so it is fixed. A teacher who signed up as a student by mistake is deleted (while unused) and recreated as a teacher.

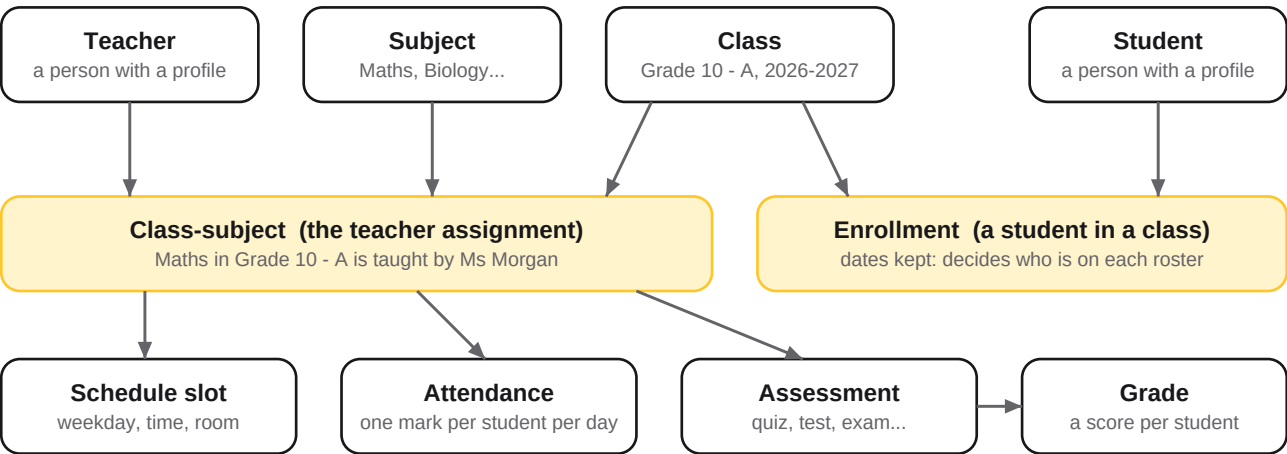
### What happens when you sign in



This runs on **every request**, so a deactivated person is locked out on their very next click.

- **Forgot password** emails a reset **link** (not a code). The same message appears whether or not the email exists.
- Demo accounts use fake addresses and cannot receive email. To try it, register a student with a real address.
- **Deactivate** blocks access and keeps all records (a student's active enrollment is withdrawn). **Delete** is only for an unused, mistaken account.

## 4. How a school is organised



| Concept                | What it is, and the rules                                                                                                                                                                                  |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Subject and class      | A subject is taught (Maths). A class is a group for one school year, with a grade level and an optional homeroom teacher. The year starts in <b>August</b> (2026-2027). Anything in use cannot be deleted. |
| Class-subject          | A subject in a class taught by one teacher. It anchors the timetable, attendance and grades. Changing the teacher keeps earlier records.                                                                   |
| Enrollment             | A student in a class, with dates. <b>One active enrollment at a time.</b> A transfer closes the old one and opens a new one, so history is never overwritten.                                              |
| Timetable              | Weekly slots (day, time, room). Refused if the <b>class, teacher or room</b> is already booked; the error lists the clashing lessons.                                                                      |
| Attendance             | Per lesson and day: <b>present, absent, late or excused</b> . The roster is the students who were in the class that day. No future dates. <b>Rate = (present + late) / marked.</b>                         |
| Assessments and grades | A quiz, test or exam with a maximum score. A score must be 0 to the maximum. Results are <b>points-weighted</b> (total scored / total possible), so bigger tests count more.                               |
| Announcements          | Audience (everyone, teachers or students), optional class, optional publish and expiry dates. Teachers post only to their own classes and edit only their own posts.                                       |

## 5. A day in the life

| Administrator                                                                                                                                                                                                                                                                      | Teacher                                                                                                                                                                                                                                         | Student                                                                                                                                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ol style="list-style-type: none"> <li>1. Create teachers and students</li> <li>2. Create subjects and classes</li> <li>3. Assign a teacher to each subject</li> <li>4. Enroll students</li> <li>5. Build the timetable</li> <li>6. Watch the dashboard, handle changes</li> </ol> | <ol style="list-style-type: none"> <li>1. Read the Today dashboard</li> <li>2. Mark attendance</li> <li>3. Create an assessment, enter the scores</li> <li>4. Post an announcement to a class</li> <li>5. Check the weekly timetable</li> </ol> | <ol style="list-style-type: none"> <li>1. See today's timetable</li> <li>2. Check attendance rate</li> <li>3. Read grades per subject</li> <li>4. Read announcements</li> </ol> |

**Dashboards differ by role.** The administrator sees counts, attendance today and upcoming assessments; the teacher sees today's periods and pending grading; the student sees their timetable, attendance and latest grades. Every list has search, filters, sorting and paging, and a light or dark theme switch. On phones the menu becomes a bottom bar.

## 6. Safety and rules

- Every request is checked for a real sign-in, an active account and the right role. The API enforces it even if a screen is bypassed.
- Input is validated strictly, SQL is parameterised, and only the allowed websites may call the API from a browser.
- The database itself enforces key rules (unique emails, one active enrollment, valid scores), so a code bug cannot break them.
- A refused action shows a clear message: 401 not signed in, 403 not allowed, 404 not found, 409 conflict (duplicate, in use, clash), 400 invalid input.

## 7. Hosting, data and limits

| What     | Where                              | Good to know                                                                     |
|----------|------------------------------------|----------------------------------------------------------------------------------|
| Website  | skoleph.web.app (Firebase Hosting) | Free, always on                                                                  |
| API      | skole-api.onrender.com (Render)    | Free; sleeps after 15 idle minutes, the next request takes about a minute        |
| Database | Aiven MySQL free plan              | Free, 1 GB; may be switched off after long inactivity (turn it back on in Aiven) |

**Where the data lives.** Nothing is stored on anyone's computer, so changing laptops loses nothing. The browser keeps only the signed-in session and the theme choice.

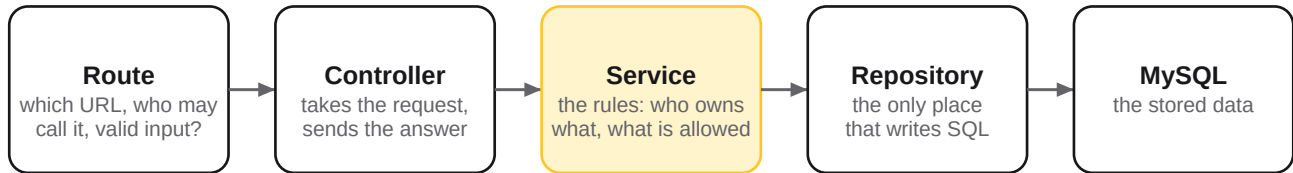
**Real time?** Everyone shares the same data and every save is written at once. Open pages do **not** update by themselves: they show other people's changes the next time the page loads (data counts as fresh for 30 seconds). If two people edit the same record, the last save wins.

**Run it locally:** the README lists every step (Node 24, MySQL 8, a free Firebase project, then `npm run db:migrate`, `npm run db:seed` and `npm run dev` in backend and frontend).

**Demo accounts** (password Password123!): admin@school.test and admin2@school.test, teacher1 to teacher3, student1 to student8, all @school.test.

## 8. Under the hood

React 19 and Tailwind on the website, Node.js 24 and Express 5 for the API, MySQL 8 with 12 tables, Firebase Authentication, 64 documented API endpoints and 128 automated tests. Every feature follows the same path:



The **service** holds the rules and the **repository** is the only place that writes SQL, so a bug is found by following one path. The website hides what you cannot do, but the API is what enforces it.

## 9. Quick answers

| Question                             | Answer                                                                   |
|--------------------------------------|--------------------------------------------------------------------------|
| Can there be only one administrator? | No, as many as needed. At least one active administrator always remains. |
| Is the administrator above everyone? | Yes. They manage accounts and all school data.                           |
| How does a teacher get an account?   | An administrator creates it and gives the teacher a temporary password.  |
| Can a student become a teacher?      | No. Roles are fixed. Delete the unused account and create a new one.     |
| What if a student changes class?     | The administrator transfers them. The old enrollment stays as history.   |
| Does deactivating erase data?        | No. Records stay. The person simply cannot sign in until reactivated.    |
| Why was the first page slow?         | The free server sleeps when idle and wakes on the first request.         |